Project Report

AI-Driven Attack Defense for IoT Smart City in 6G Networks

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Chapter 1

Scientific Context

Smart Cities represent a major evolution in modern urban development, aiming to improve citizens' quality of life through the intelligent use of digital technologies. By integrating ICT, IoT, cloud/edge computing, big data analytics, and AI, Smart Cities enable more efficient management of urban resources and services (transportation, energy, safety, healthcare, environment, etc.).

IoT devices such as sensors, cameras, smart meters, and connected vehicles continuously generate massive volumes of heterogeneous data. These data are transmitted over advanced communication infrastructures, including emerging sixth-generation (6G) networks, which target ultra-low latency, high data rates, massive connectivity, and stronger AI-native automation.

However, deploying IoT at city scale significantly increases the attack surface. Cyberattacks targeting IoT and network infrastructures can disrupt critical services, compromise data, and threaten public safety. Ensuring cybersecurity in Smart City environments is therefore a major scientific and technological challenge.

The convergence of IoT, 6G networks, and AI-driven analytics accelerates Smart City innovation while introducing complex cybersecurity challenges that require intelligent, automated, scalable, and near real-time detection and defense mechanisms.

1.1 How a Smart City Works (system view)

Figure 1.1 illustrates a four-layer smart city architecture adapted from Zhao and Zhang (2020). The architecture is structured as follows: (i) the **Perceptual layer**, which includes sensors, cameras, user devices, and intelligent terminals that collect real-world data; (ii) the **Network layer**, responsible for transmitting data through communication networks and the Internet; (iii) the **Platform layer**, which manages data storage, integration, standards, and modeling; and (iv) the **Application layer**, where smart services

such as smart government, transportation, healthcare, and safe city systems are delivered.

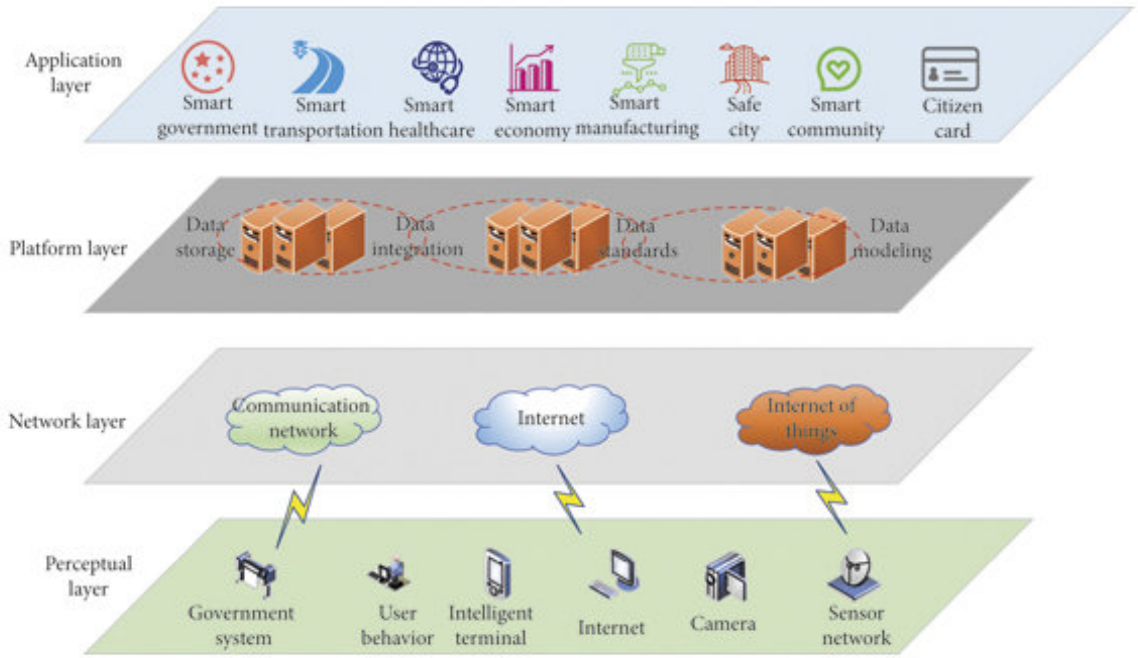

Figure 1.1: Four-layer Smart City architecture (Perceptual, Network, Platform, and Application layers). Adapted from: Zhao, Z., & Zhang, Y. (2020). *Impact of Smart City Planning and Construction on Economic and Social Benefits Based on Big Data Analysis.* Complexity, 2020, 1–11.

1.2 Comparison of 5G and 6G Networks for Smart Cities

Figure 1.2 illustrates the architectural evolution from 5G to 6G networks, adapted from Saad et al. (2020). In the 5G architecture, the network is structured into user, access, and core planes, relying mainly on centralized cloud and IP computing, SDN/NFV, and network slicing to support enhanced mobile broadband and IoT services.

In contrast, the 6G architecture introduces an **AI-native and intelligent end-to-end design**. It integrates edge intelligence (MEC), fog computing, intelligent access networks, and intelligent core management. 6G extends connectivity beyond traditional mobile devices to include industrial IoT, healthcare IoT, autonomous vehicles, drones, and massive machine-type communications. This shift enables ultra-low latency, extreme automation, and real-time intelligent decision-making, which are essential for large-scale Smart City environments.

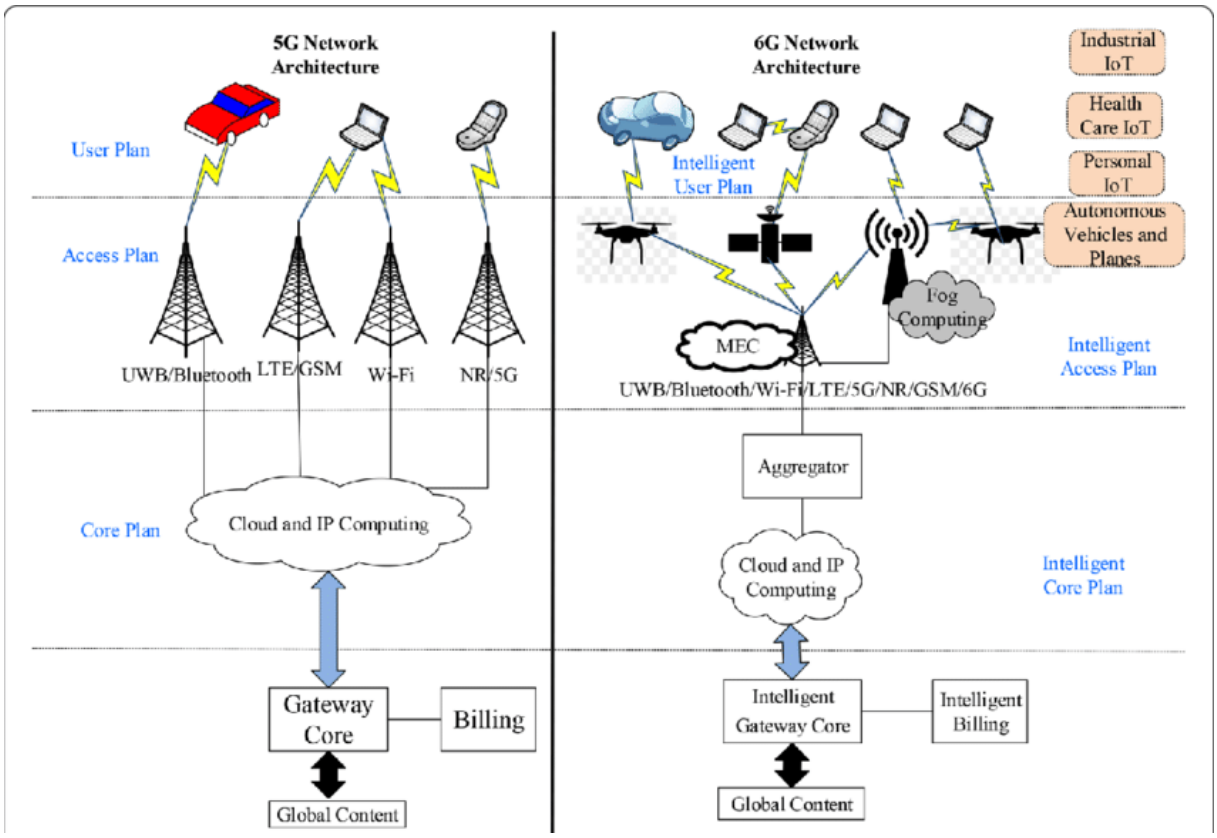

Figure 1.2: Comparative architecture of 5G and 6G networks. 6G introduces intelligent user, access, and core planes with integrated AI, edge (MEC), and fog computing. Adapted from: Saad, W., Bennis, M., and Chen, M. (2020). *A Vision of 6G Wireless Systems: Applications, Trends, Technologies, and Open Research Problems.* IEEE Network.

Chapter 2

Problem Formulation: Study & Critique of Existing Approaches

2.1 Central Problem Statement

Smart City networks are highly distributed and heterogeneous, combining large numbers of IoT devices (often resource-constrained), mobile users, and latency-critical services. In 5G/6G-style deployments, traffic is also separated into service contexts/slices (e.g., eMBB, mMTC, URLLC), where "normal" behavior differs significantly across services.

In this setting, attacks such as DDoS/botnets, scanning, spoofing, malware-driven abnormal traffic, and resource exhaustion can emerge at multiple entry points and propagate quickly. Classic defenses that rely on static rules, signatures, or centralized analysis struggle to:

- **Keep up with scale:** massive numbers of flows and continuous traffic make heavy-weight inspection difficult.
- **Handle evolving threats:** variants and previously unseen device behaviors reduce the effectiveness of signature-only methods.
- **Operate close to real time:** low-latency services (URLLC) require fast detection decisions with minimal overhead.
- **Work under heterogeneous traffic:** different access locations and device clusters create non-uniform (non-IID) traffic distributions, making a single global detector unreliable at all points.

How can we design a flow-based, AI-driven detection system for Smart City traffic over 5G/6G-style infrastructures that (i) detects attacks including variants, (ii) remains scalable for high-throughput user-plane monitoring, (iii) is robust to heterogeneous traffic across locations and services, and (iv) supports near real-time operation with low false-alarm rates?

2.2 Where the Threats Appear in a 6G/5G Smart City (attack surface)

- **IoT/UE layer:** compromised devices generating botnet/DDoS traffic, abnormal scanning behavior, credential abuse.
- **Access / entrance points:** large volumes of IoT traffic enter through multiple access locations; distributed attacks may be visible only when considering multiple points together.
- **Edge/MEC:** attacks against edge services (IoT brokers, analytics), resource exhaustion, poisoning of local monitoring/analytics pipelines.
- **Core user-plane (UPF/N6) and service functions:** abuse of user-plane capacity, abnormal flow patterns, and service/API misuse.
- **Slicing/virtualization:** slice resource exhaustion, misconfiguration risks, and isolation failures that can amplify the impact of abnormal traffic.

2.3 Study and Critique of Existing Approaches

A) Signature / Rule-Based IDS/IPS and Firewall Policies

What they propose: detect/block threats using known signatures, rules, and policy constraints (e.g., allow/deny lists, thresholds).
Strengths: fast, interpretable, effective for known threats and clear policy violations.
Limits: limited ability to detect new/variant behaviors; policies become complex at scale; performance may degrade when traffic is high and heterogeneous.

B) Payload Inspection and DPI-Based Security (NGFW-style)

What they propose: deep inspection to identify applications and known malicious payload patterns.

Strengths: strong visibility when payload is available.
Limits: encryption reduces payload visibility; inspection cost can be high for large-scale Smart City traffic; not always feasible at edge/high-throughput points.

C) Centralized ML/DL-Based Intrusion Detection

What they propose: train ML/DL models centrally (cloud/SOC) using aggregated traffic data to classify malicious vs benign flows.
Strengths: can learn behavioral patterns beyond signatures; strong accuracy in controlled datasets.
Limits (in distributed Smart City settings):

- requires moving large volumes of data to a central location (bandwidth and privacy concerns),
- a single centralized model may become heavy for deployment at constrained entrance/edge points,
- generalization drops when traffic distributions differ strongly across locations and device clusters.

D) Distributed Detection with Federated Learning (FL)

What it proposes: keep data local at detection points and collaborate by sharing model updates/parameters instead of raw traffic.
Strengths: reduces data sharing and can improve coverage across distributed points.
Key limitation: standard FL typically produces a *global averaged model* that may not perform best at each local point when traffic is heterogeneous (non-IID), leading to weaker local detection.

E) Access-Side Distributed Detection with Personalized FL

Recent research by **Zhou et al.** proposes shifting detection **forward to the access side** (entrance points) using a **transformer-based model** and a **personalized federated learning** collaboration mechanism. The idea is to allow multiple distributed detection points to share model parameters while still keeping **personalized layers** optimized for local traffic, improving detection under heterogeneous distributions and supporting detection of traffic from previously unseen device types at local points.
Strengths:

- addresses distributed entry-point visibility via multi-point collaboration,
- aims to reduce the resource bottleneck of purely victim-side/centralized detection,

- explicitly targets heterogeneous IoT traffic distributions across detection points.

Practical critique / remaining gaps for Smart City 5G/6G deployments:

- **Operational constraints:** as the number of edge points grows, frequent parameter aggregation can reduce update speed, affecting responsiveness.
- **Scope of attacks and environments:** performance is typically demonstrated on specific IoT botnet/DDoS datasets; broader Smart City traffic and multi-service contexts may require additional validation.
- **Actionability:** high accuracy is not sufficient alone; operators need prioritized alerts and clear reasons to reduce false-alarm burden.

2.4 Resulting Gap and Project Direction

From the critique above, the key gap is a deployable detection approach that is **behavior-based, scalable, and operationally usable** for Smart City user-plane monitoring. Therefore, this project focuses on:

- **Flow-based detection at user-plane monitoring points (UPF/MEC):** use statistical flow features (rates, timing, flags, directionality) that remain available even when payload is encrypted.
- **Context-aware detection across services (eMBB/mMTC/URLLC):** reduce false alarms by modeling "normal" behavior per service context and detecting deviations/attacks relative to that context.
- **Operational outputs:** produce risk scoring and prioritized alerts with simple feature-based explanations to support faster response.
- **Robustness over time:** monitor data drift and update the model as Smart City traffic evolves.

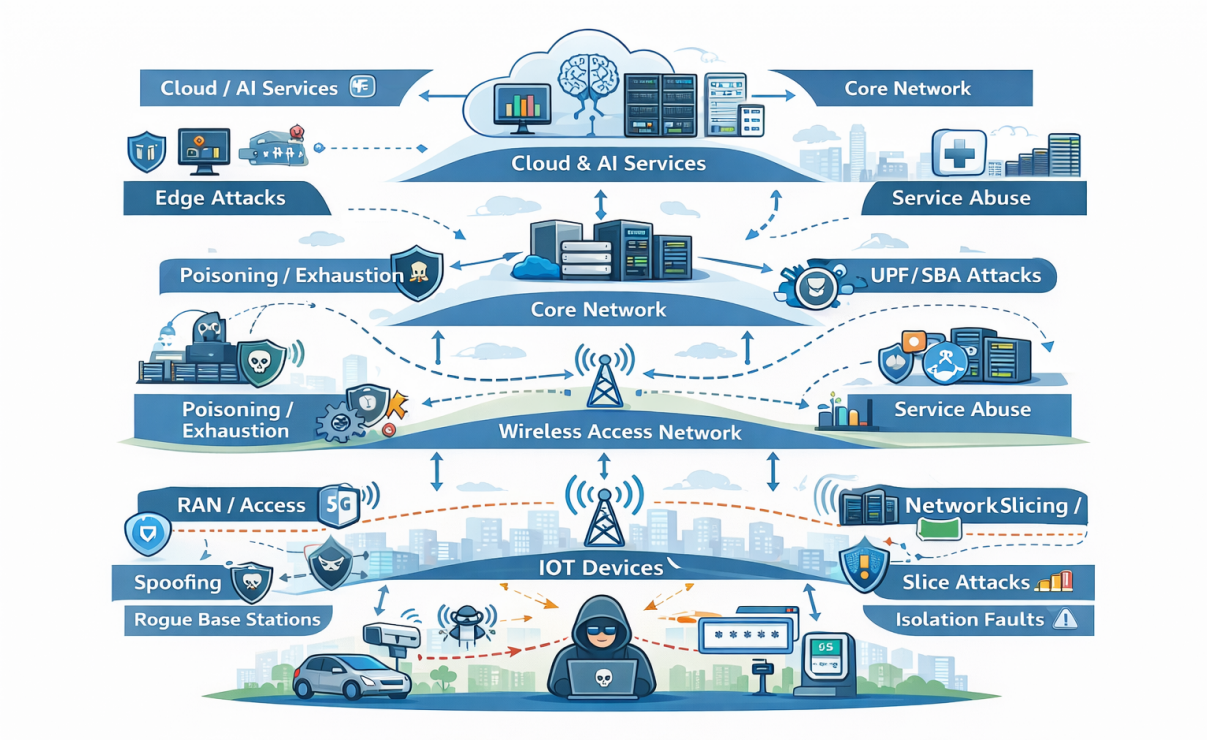

Figure 2.1: Common vulnerabilities and cyber-attack types in IoT-based Smart City networks.

Chapter 3

Business Objectives

This project aims to protect Smart City digital services (transportation, public safety, energy, healthcare, and environment monitoring) from cyber threats by reducing successful attacks, accelerating response, and lowering the day-to-day operational effort needed to stay protected as services scale.

BO0 – Protect priority clients during cyber incidents

Goal: Ensure that high-priority Smart City clients (such as healthcare institutions, emergency services, and public safety authorities) remain fully operational and protected, even during cyber-attacks or abnormal network conditions.

KPIs:

- Availability rate for priority clients during incident scenarios (target: maintain $>$ 99.9%).
- Percentage of priority clients unaffected during network attacks (target: increase).
- Performance degradation experienced by priority clients under abnormal traffic (target: decrease).
- Average time to fully restore service quality for priority clients after disruption (target: decrease).

BO2 — Reduce false alarms and speed up operator response

Goal: Provide clear, prioritized early warnings that operators can trust, enabling faster actions while minimizing time wasted on false alarms.

KPIs:

- Operator false-alarm rate (target: decrease vs baseline).
- Average time between an early warning and operator action (target: decrease).

- Percentage of warnings that are confirmed as real issues after review (target: increase).

BO3 — Lower operating costs while staying ready as the city grows

Goal: Protect the network at large scale without heavy manual work, and keep protection effective as services expand and usage patterns change.
KPIs:

- Reduction in monthly human effort spent on manual updates and investigations (hours/month).
- Time required to roll out an improvement to detection logic (target: decrease).
- Coverage of monitored services and connections (target: increase while maintaining stable operations).

Chapter 4

Data Science Objectives

Each Business Objective is translated into one or more Data Science Objectives to ensure traceability between stakeholder value and technical work.

BO1 – Stop More Attacks Before They Cause Service Disruption

- **DS1.1:** Train a behavior-based detection model using flow features (timing, rates, direction, TCP flags) to detect attacks and variants.
- **DS1.2:** Enable early detection by predicting attack risk from short observation windows (first part of a flow).
- **DS1.3:** Make detection context-aware by separating traffic into service contexts (eMBB/mMTC/URLLLC) and applying specialized detection per context.

BO2 – Reduce False Alarms and Speed Up Operator Response

- **DS2.1:** Reduce false alarms by tuning the model for precision while maintaining strong attack recall on imbalanced data.
- **DS2.2:** Produce a calibrated risk score and alert severity levels (high/medium/low) to prioritize operator actions.
- **DS2.3:** Provide simple alert explanations by returning the top contributing flow features for each prediction.

BO3 – Lower Operating Costs While Staying Future-Ready

- **DS3.1:** Deploy a lightweight, scalable inference pipeline based on flow statistics suitable for high-throughput environments (UPF/MEC).
- **DS3.2:** Monitor data drift and performance drift to detect when traffic behavior changes and retraining is needed.

- **DS3.3:** Version datasets and models and compare candidates against a baseline to safely update detection over time.

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Chapter 5

SDG Objectives

This project contributes to the United Nations Sustainable Development Goals (SDGs) by strengthening the security and reliability of digital infrastructures that support Smart City services. Cybersecurity is a key enabler of safe, resilient, and trusted urban systems.

5.1 Relevant SDGs and Alignment with the Project

SDG 8 – Decent Work and Economic Growth

What it means: SDG 8 promotes sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work. Reliable digital infrastructures are essential to support modern economic activities and smart urban services.
Alignment with our project:

- Cyber-attacks against Smart City networks can cause service outages and significant economic losses; our project helps reduce these risks by enabling **early detection and mitigation**.
- By improving the reliability and availability of IoT-based services, the solution supports **continuous economic activity** in smart urban environments.
- A secure digital infrastructure reduces operational disruptions and contributes to a more **stable and resilient digital economy**.

SDG 9 – Industry, Innovation and Infrastructure

What it means: SDG 9 focuses on building resilient infrastructure, promoting inclusive and sustainable industrialization, and fostering innovation. In the Smart City context, this includes modern communication networks (5G/6G), digital public services, and data-driven infrastructures.
Alignment with our project:

- Our solution improves the **resilience** of communication infrastructure by detecting cyber-attacks that could disrupt connectivity.
- By enabling **automated, AI-driven defense**, the project supports innovation in next-generation network security.
- Protecting critical network services increases **infrastructure reliability** for industries and public operators that depend on digital connectivity.

SDG 11 – Sustainable Cities and Communities

What it means: SDG 11 aims to make cities inclusive, safe, resilient, and sustainable. Smart Cities rely on connected systems for transportation, energy, waste management, environmental monitoring, and public safety.
Alignment with our project:

- Cyber-attacks can stop or degrade essential urban services; our project helps ensure **service continuity** and **urban resilience**.
- By detecting malicious activity in IoT environments, we protect the digital layer that supports **sustainable city operations**.
- A reliable security layer increases citizen trust and encourages adoption of Smart City technologies that improve quality of life.

By improving cybersecurity for IoT-based Smart City services over next-generation networks, the project supports sustained economic activity (SDG 8), resilient digital infrastructure (SDG 9), and safer, more sustainable urban services (SDG 11).

The following figure summarizes the Sustainable Development Goals addressed by the project and illustrates their relevance to Smart City cybersecurity.

Figure 5.1: Sustainable Development Goals supported by the project.

Chapter 6

Data Understanding: Dataset Description

6.1 Overview

The datasets used in this project represent network communications as **bidirectional flows** (flow-based traffic analysis). A flow summarizes a communication session between two endpoints by aggregating packet-level information into statistical and temporal descriptors. This representation is widely used for intrusion detection because it reduces the complexity of packet traces while preserving patterns that differentiate normal behavior from malicious activity.

Based on the attached EDA notebooks, the project relies on two complementary data sources:

- A **6G IoT Smart City flow dataset** (high-dimensional flow features, similar to CICFlowMeter-style variables).
- Multiple **5G Smart City datasets** representing different service scenarios/slices: **eMBB**, **mMTC**, **URLLC**, and a **Global** dataset combining them.

6.2 Dataset 1: 6G IoT Smart City Flow Dataset (EDA summary)

6.2.1 Size and structure

From the notebook exploration:

- Dataset shape: **(2614, 85)** (2614 flows, 85 columns).
- Columns include identifiers (Flow ID, IPs, ports, timestamps) and flow statistics.

- After removing a subset of identifiers (Flow ID, Source IP, Destination IP, Timestamp, External IP), the notebook reports **79 numeric features** used for modeling.

6.2.2 Feature categories (observed)

The 6G dataset contains several families of features:

- **Identifiers and context:** Flow ID, Source/Destination IP, Source/Destination Port, Protocol, Timestamp, External IP.
- **Volume and direction statistics:** total forward/backward packets and bytes; forward/backward packet length min/max/mean/std.
- **Rates:** Flow Bytes/s, Flow Packets/s, forward/backward packets per second.
- **Temporal dynamics (IAT):** Flow IAT and direction-specific IAT (mean/std/min/max/total).
- **TCP behavior:** flag counters (e.g., SYN/ACK/RST/FIN) and header-related features (when available).
- **Activity/Idle behavior:** Active and Idle min/mean/max/std (useful for bursty traffic and periodic IoT patterns).

6.2.3 Data quality and missing values

The notebook reports **very low missingness** in the 6G dataset: **68 columns** have missing values, but only **1 missing value per affected feature**. This corresponds to a negligible missing rate (approximately $1/2614 \approx 0.038\%$ per feature), meaning simple strategies (row removal or light imputation) are sufficient.

6.2.4 Label distribution (important observation)

In the notebook's label distribution output, the `Label` column appears to be dominated by a single class (shown as **Benign = 100%** in the printed proportions). This indicates one of the following:

- the dataset subset used in the notebook contains mostly/only benign flows, or
- labels require verification (encoding issue, filtering step, or missing attack categories).

This is a critical point because supervised classification requires multiple classes. If labels are effectively single-class, the dataset becomes better suited to **anomaly detection**, **self-supervised learning**, or must be complemented with additional labeled attack data.

6.2.5 Feature redundancy (correlation findings)

The 6G notebook highlights many **highly correlated or nearly duplicate features** (correlation ≥ 0.9), for example:

- Avg segment size vs packet length mean (correlation near 1.0),
- Activity/Idle indicators closely related to IAT extremes,
- Multiple packet-length statistics strongly correlated with each other.

This redundancy motivates dimensionality reduction or feature selection to improve generalization and reduce inference cost.

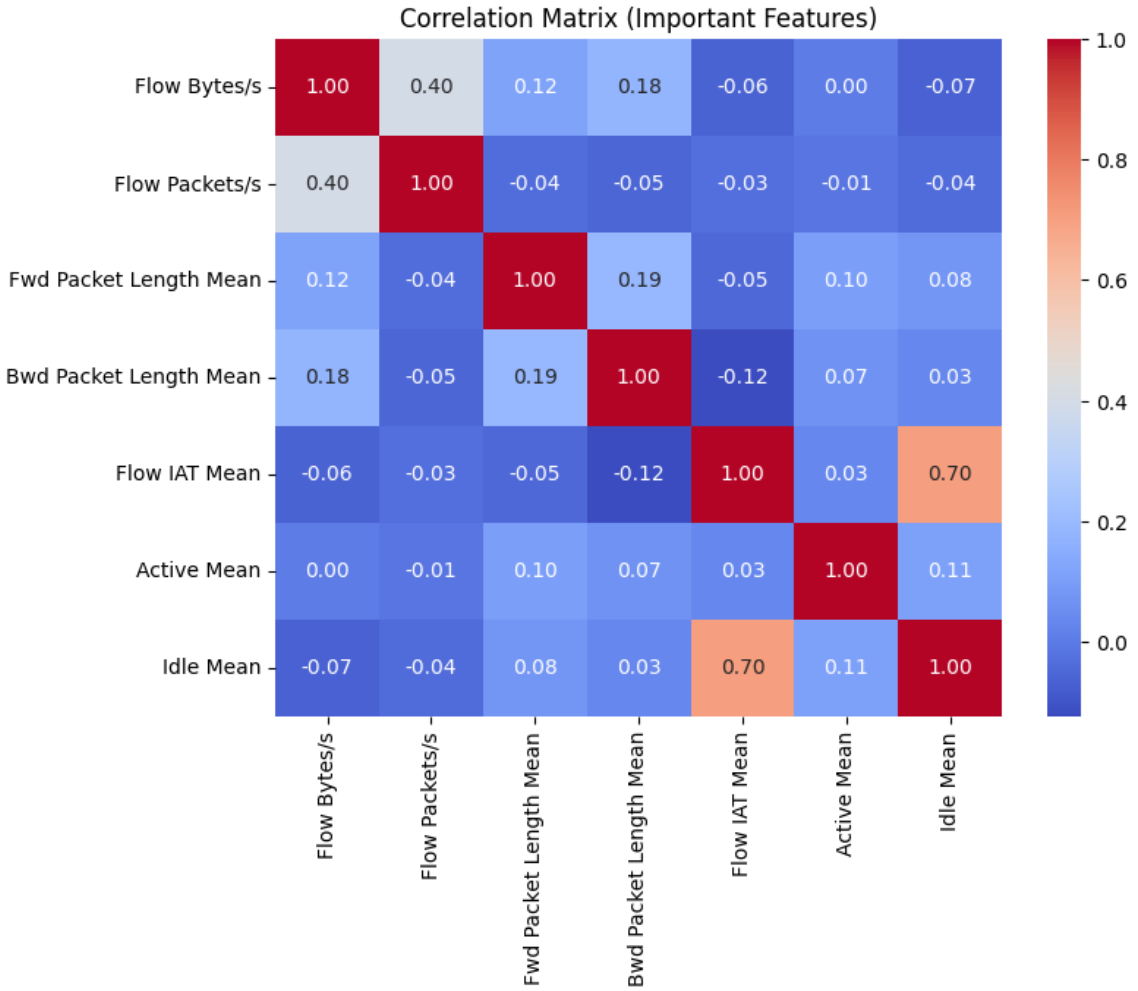

Figure 6.1: Correlation heatmap for selected important features in the 6G dataset (EDA output).

6.2.6 Baseline importance signals (from notebook)

A quick Random Forest baseline in the notebook suggests that flow duration and packet-length-related features are among the most influential features for discrimination. However, the notebook also shows `Source Port` ranking high, which is often a **data leakage**

risk (models can memorize scenario-specific ports rather than learning behavioral patterns). For robust generalization, port/IP identifiers should be excluded or carefully handled (e.g., aggregated statistics rather than raw values).

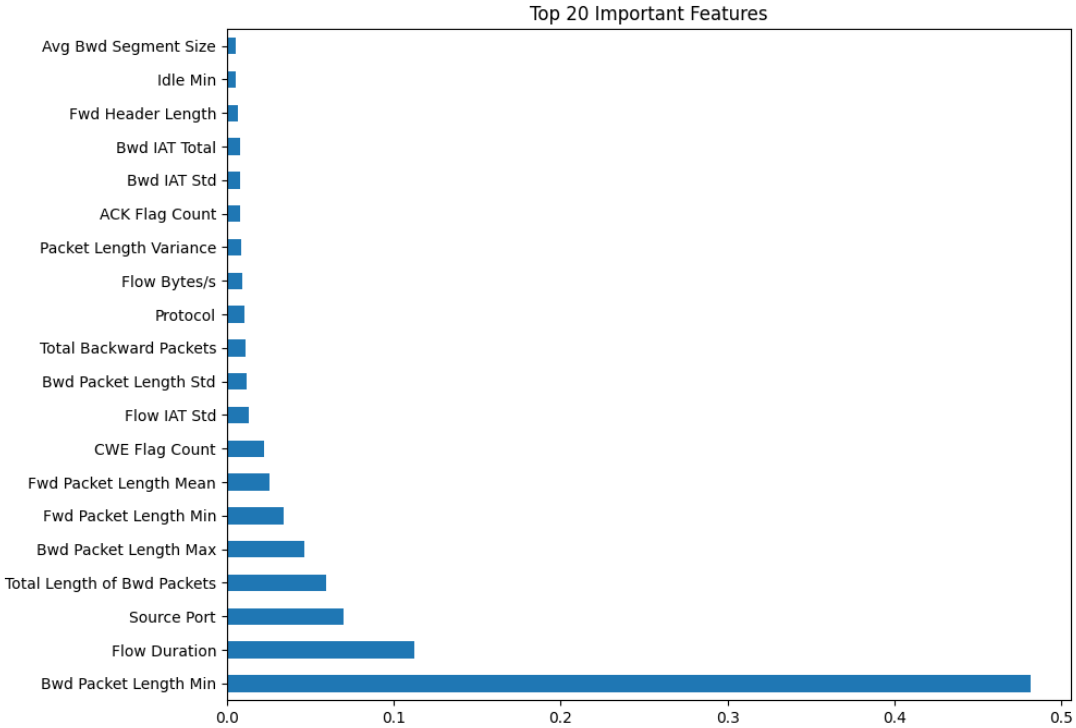

Figure 6.2: Top 20 features from a Random Forest baseline (6G dataset, EDA output).

6.3 Dataset 2: 5G Smart City Datasets (eMBB, mMTC, URLLC, Global)

6.3.1 Size and structure

From the 5G notebook:

- **eMBB shape: (5808, 52)**
- **mMTC shape: (4615, 52)**
- **URLLC shape: (4033, 52)**
- **Global shape: (14456, 52)**

All 5G datasets share a consistent 52-column schema, which supports cross-slice comparisons and unified preprocessing.

6.3.2 Feature categories (observed)

The 5G datasets combine:

- **Flow summary statistics:** duration (`Dur`), mean/sum/min/max, runtime.
- **Traffic volumes:** total/src/dst packets and bytes (`TotPkts, SrcPkts, DstPkts, TotBytes, SrcBytes, DstBytes`).
- **Protocol and network fields:** protocol (`Proto`), ToS/TTL/Hops (`sTos/dTos, sTtl/dTtl, sHops/dHops`).
- **Performance/behavior indicators:** rate and loads (`Rate, SrcRate, DstRate, Load, SrcLoad, DstLoad`), loss (`Loss, SrcLoss, DstLoss, pLoss`).
- **TCP timing/flags-like indicators:** `TcpRtt, SynAck, AckDat`.
- **State/context fields:** `Cause, State` (categorical).
- **Target fields:** `Label` and a `predicted` column (present in the dataset as provided).

6.3.3 Missing values (key findings)

The 5G notebook reports **significant missingness in specific fields**, which varies by slice:

- In **eMBB**, `sVid` and `dVid` are fully missing (**5808 missing each**), meaning these variables carry no information for that dataset and should be dropped.
- In **mMTC** and **URLLC**, several network-related fields contain missing values (e.g., `dTos, dTtl, dHops, DstWin, SrcTCPBase, DstTCPBase`).
- In the **Global** dataset, `sVid` and `dVid` are almost entirely missing (**14082** and **14452** missing), strongly suggesting removal.

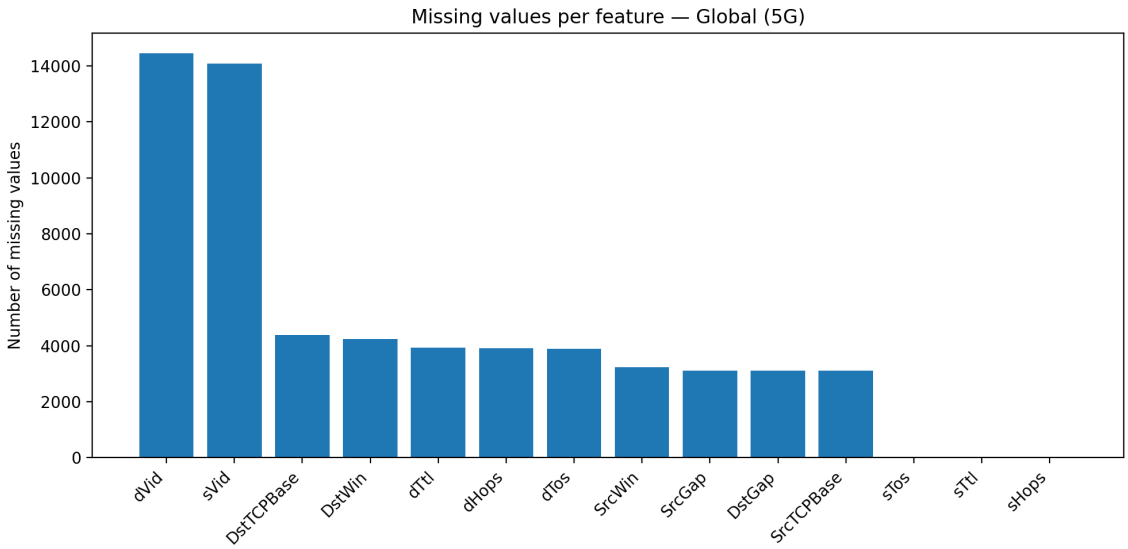

Figure 6.3: Missing values per feature in the Global 5G dataset (EDA output).

6.3.4 Label distribution (important observation)

In the notebook's "normal vs attack" split for the Global dataset, the reported counts are: **Normal flows: 0** and **Attack flows: 14456**. This indicates an **extreme imbalance or a label-encoding issue** (e.g., all samples assigned to a single class, or normal/attack values inverted). This must be validated before supervised modeling. If only one class is present, the dataset is not directly usable for binary supervised classification and would require:

- label verification/correction, or
- adding normal samples / additional sources, or
- switching to anomaly detection or one-class methods.

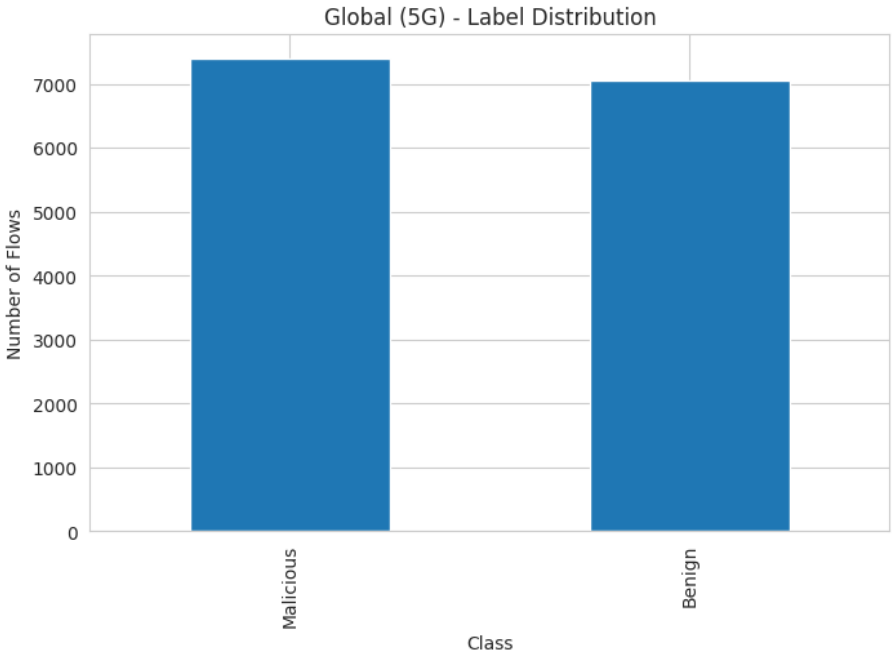

Figure 6.4: Label distribution in the Global 5G dataset (EDA output).

6.4 Data challenges and implications for modeling

The EDA notebooks highlight several practical challenges:

- **High dimensionality and redundancy** (especially in the 6G dataset) ⇒ requires feature selection/reduction.
- **Missing values** (especially in 5G datasets) ⇒ requires dropping unusable columns and imputing partially missing fields.
- **Label imbalance / potential single-class labeling** ⇒ impacts supervised training and requires careful validation and alternative strategies.
- **Risk of data leakage** from identifiers (IPs/ports, scenario artifacts) ⇒ encourages behavioral feature focus rather than raw identifiers.

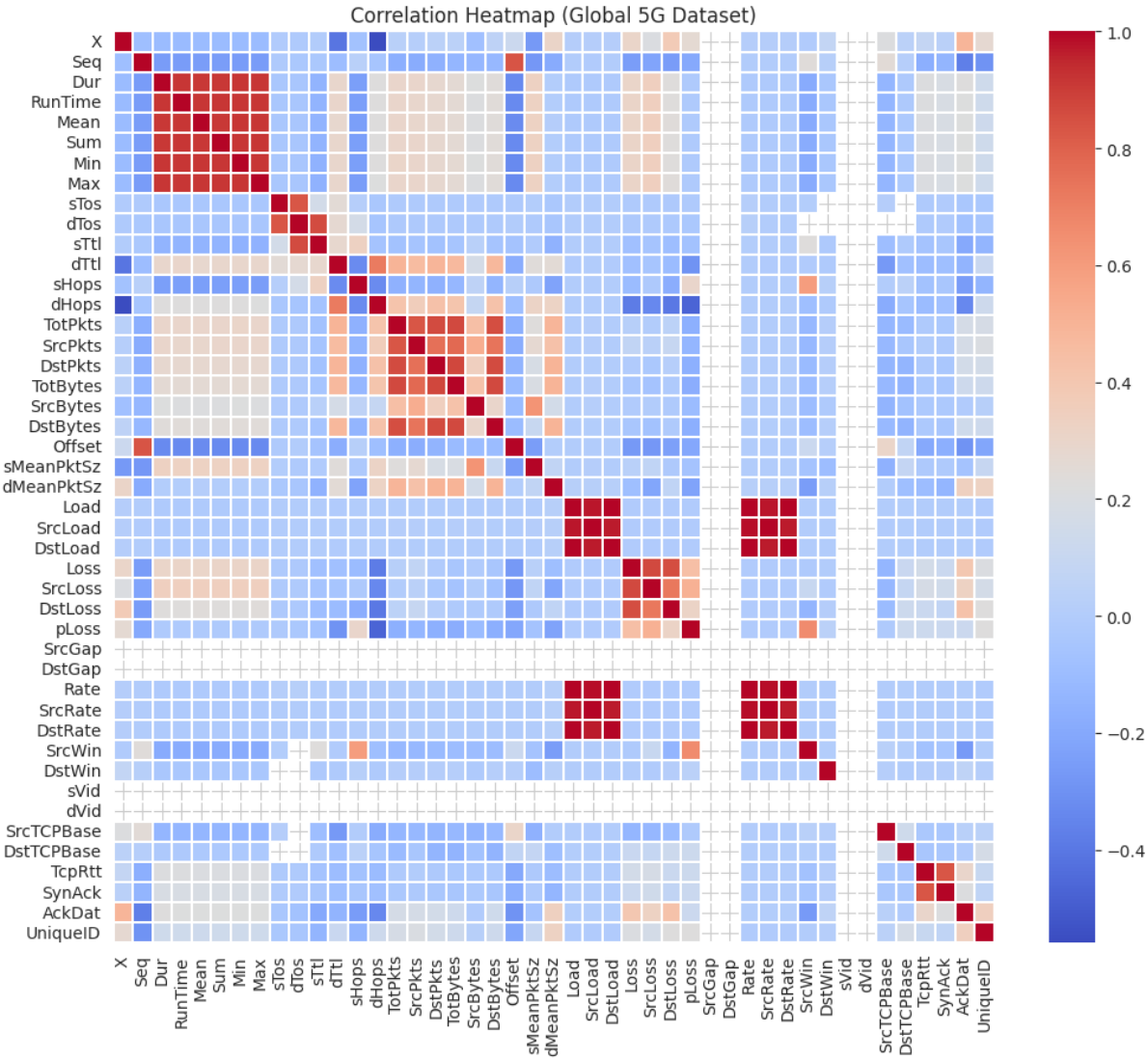

Figure 6.5: Correlation heatmap of numeric features in the Global 5G dataset (EDA output).

6.5 Recommended preprocessing plan (derived from EDA)

- **Identifier handling:** remove or strictly control IP/port/timestamp usage (avoid memorization; prefer behavioral statistics).
- **Missing values:** drop fully-missing fields (e.g., `sVid/dVid`), impute partial missingness (median/most frequent).
- **Categorical encoding:** encode `Proto, Cause, State` using one-hot encoding or label encoding (depending on model choice).
- **Scaling:** normalize heavy-tailed variables (rates, bytes/s) using log-transform + standardization when appropriate.

- **Redundancy reduction:** remove highly correlated features (e.g., correlation > 0.95) or use PCA/feature selection.
- **Validation strategy:** adopt stratified splits when labels are valid; consider time-based validation for realistic deployment.

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